Lab 2.3 Risk assessment and method statement with EMAP

IES | 20 minutes | LO2, LO3

All records are fictional training data. They describe no actual facility, person or organisation. Values are teaching assumptions, not approved design criteria. Do not use outputs to operate, maintain or approve real equipment.

Inputs and preparation

Open the mock-data folder: work-activity.txt, hazard-checklist.csv, security-checklist.csv. Use prompts.md for copy and paste. Save your reviewed work under outputs. Earlier lab outputs are optional; this folder provides standalone inputs.

Procedure

1. Read work-activity.txt. This is a desk exercise: no physical task may be performed. Identify missing chemical identity, SDS and isolation information.

2. In Word, build an E-MAP prompt: engineering context, measurable review criteria, explicit assumptions, and parameters or boundaries. Use the prompt below.

3. Generate a planning risk table with hazard, affected person, consequence, existing evidence, proposed controls, missing information and reviewer. Use qualitative risk discussion unless an approved rating matrix is supplied.

Lab 2.3 continued

4. Generate a method-statement planning outline containing prerequisites, hold points, references to approved site procedures, emergency planning and handback. Reject invented operational isolation or PPE instructions.

5. Use hazard-checklist.csv to challenge the draft. Ask for missed hazards and contradictory assumptions. Save both draft versions and an unresolved-items list; mark release blocked pending competent review.

Check before moving on

All seven supplied hazard categories are addressed.

Missing SDS and isolation information block release.

Risk controls are proposals and not approved instructions.

Access or source problems

Use approved paired access or the local source files. Record unavailable live capabilities as not executed. If an attachment is not supported, paste the short synthetic text or reference its approved SharePoint copy. Do not use personal accounts to bypass controls.

Lab 2.3 prompts and evidence

Main prompt

Engineering: planning-stage replacement of the fictional dosing pump. Metrics: cover every hazard in hazard-checklist.csv and expose all missing evidence. Assumptions: no chemical identity, SDS or approved isolation details are available. Parameters: desk exercise only; no work is authorised. Draft a risk assessment and method-statement planning outline. Include hold points and competent review. Do not prescribe chemical-specific PPE, exposure limits or live-work instructions.

Verification prompt

Audit the draft against the named source files. List each claim, source location, uncertainty and correction. Recalculate key values independently. Return a revised draft and change log; do not approve the engineering deliverable.

Expected output example

Illustrative teaching example, not a recorded AI run. Chemical exposure: substance unidentified; SDS required; chemical risk reviewer to confirm controls. Hold point: do not proceed to work planning approval until identity and compatibility evidence are available.

Retain your evidence

Keep the source versions, prompt, tool, output, corrections, independent checks and reviewer decision. Record any capability or permission test that was not executed. Use templates/prompt-log.csv. Do not publish real workplace data.